

## RESEARCH PROFILE

**Seeking a funded Doctoral Researcher or Research Assistant position in HCI, human-AI interaction, or immersive systems, with availability in 2027.**

HCI researcher building and evaluating interactive AI systems in AR/XR. Uses controlled experiments and interaction logs to study how people interpret, reshape, adopt, or reject AI contributions, distinguishing changes in system output from changes in human thinking, agency, and experience.

## RESEARCH AND PROFESSIONAL EXPERIENCE

AXIS Lab, Interaction Science Institute, Sungkyunkwan University 2024–present  
Undergraduate Research Assistant (2024–Feb 2025); Master's Student Researcher (Mar 2025–Aug 2026); Researcher (Sept 2026–present). Contributing to projects on proactive AI and human cognitive mechanisms in always-on AI environments, and continuing prototype-based research on AR smart glasses, socially visible sensing, and immersive human-AI interaction.

Researcher and App Developer, Personalized AI Story Generation Project Apr 2026 – present  
Developing Phodong, a production-stage SwiftUI/Firebase system that turns children's real-world object discoveries into personalised stories. Includes age-adaptive generation, parent-gated controls, and an instrumented discovery-to-story pipeline for a planned family study; supported by Seoul RISE and led by Prof. Changjun Lee.

Assistant Instructor, Generative AI Content Creation Workshop, Hyundai Futurenet Jul 2026  
Supported hands-on workshop activities on generative-AI content creation, including participant guidance and practical exercises for producing AI-assisted visual and interactive media.

Research Intern, RGB Makers Co., Ltd. Aug 2025  
Analysed first-person interaction data and privacy risks in metahuman-based content, developed a MetaHuman character in Unreal Engine 5, and proposed outdoor interaction scenarios for user-centred immersive experiences.

BK21 Research Assistant, Social Interaction Big Data Education and Research Group Mar – Jun 2025  
Supported immersive-device user research at AXIS Lab through literature review, study and protocol preparation, participant recruitment materials, lab setup, research ethics documentation, and data management.

## EDUCATION

Sungkyunkwan University · Master of Science in Engineering, Immersive Media Engineering Mar 2025 – Aug 2026  
Advisor: Prof. Changjun Lee · AXIS Lab · Cumulative GPA: 4.45/4.50 (99.5/100)  
Thesis: *When Sensing Becomes Socially Visible: Wearer Meaning and Adoption Resistance in AR Smart Glasses*

Sungkyunkwan University · Bachelor of Culture & Technology; Bachelor of Economics Feb 2021 – Feb 2025  
Major in Culture and Technology Convergence; International Trade Linked Major, Economics Track. Cumulative GPA: 4.32/4.50 (98.2/100). Graduated with SKKU Summa Cum Laude.

## PUBLICATIONS

### PEER-REVIEWED CONFERENCE PUBLICATIONS

**Kim, G.,** Choi, Y., Kim, Y., & Lee, C. (2026). No Time to Say Goodbye: Emotional Loss Responses to Sudden Termination in Immersive AI Interactions. *CHI EA '26: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. ACM. DOI: [10.1145/3772363.3798687](https://doi.org/10.1145/3772363.3798687)

**Kim, G.,** Choi, Y., & Lee, C. (2025). See, Sense, Spark: Profiling Granularity in Proactive AI for Creative Research Ideation. *SIGGRAPH Asia 2025 Posters*. ACM. DOI: [10.1145/3757374.3771458](https://doi.org/10.1145/3757374.3771458)

**Kim, G.,** & Han, Y. (2024). Camp\_Around: An Integrated Tour Service for International Students in Korea. *HCI Korea 2024 Creative Award Proceedings*, 1224–1227.

### JOURNAL ARTICLES

**Kim, G.,** Kim, S. Y., Park, S., Park, S., Choi, Y., & Lee, C. (2025). What Makes Audiences Pay More for Immersive Cinema? A Conjoint Analysis of ScreenX Preferences and Willingness-to-Pay. *International Telecommunications Policy Review*, 32(4), 1–45. DOI: [10.37793/ITPR.32.4.1](https://doi.org/10.37793/ITPR.32.4.1)

**Kim, G.,** Choi, Y., & Lee, C. (2025). The Impact of Pre-adulthood Immersive Media Experiences on Continued Use Intention in Adulthood: An Empirical Study Applying the Value-based Adoption Model and Positive Technology Readiness Index. *Information Society & Media*, 26(1), 143–183.

### MANUSCRIPTS UNDER REVIEW

**Kim, G.,** & Lee, C. (under review). Socially Visible Sensing in AR Smart Glasses: Outward Recording, Societal Concern, and Wearer Stigma. *Technology in Society*. Manuscript under review.

**Kim, G.,** Choi, Y., & Lee, C. (under review). How Much Should Proactive AI Know About Its User? Profile Granularity in AI-Glasses Support for Research Ideation. *CHI 2027*. Manuscript under review.

**Kim, G.,** Choi, Y., Kim, Y., & Lee, C. (under review). Goodbyes Matter: Immersive Relational Offboarding for Reducing Loss When AI Companion Interactions End. *CHI 2027*. Manuscript under review.

SELECTED FUNDED RESEARCH PROJECTS

|                                                                                |                                                     |
|--------------------------------------------------------------------------------|-----------------------------------------------------|
| Youth-Focused AI Risk Evaluation Benchmark Development                         | Researcher · ETRI · 2026                            |
| Proactive AI and Human Cognitive Mechanisms in Always-On AI Environments       | Researcher · NRF · 2026–present                     |
| Personalized AI Story Generation for Children                                  | Researcher · Seoul RISE · 2026–present              |
| Interactive Experiential News Game Research and Development                    | Researcher · SKKU RISE · 2025–2026                  |
| Screen Multimodal Interaction Research · Big Data Analytics Team               | Researcher · Samsung Electronics · 2024             |
| Virtual-World Multi-Persona Tendencies and Cognitive Enhancement               | Researcher · NRF Korea · 2025                       |
| Urban Green Exposure, Mobility, and Health                                     | Researcher · AI Convergence Research Support · 2025 |
| Psychological Counselling Chatbots: Humanities–Engineering Convergence         | Researcher · NRF Korea · 2026                       |
| News and Social-Media Issue Analysis for the National Health Insurance Service | Researcher · NHIS · 2024                            |

SELECTED RESEARCH AND PROJECTS · PORTFOLIO AND DEMOS

|                                                                                                                                                                                                                                                                                                                                                                   |                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| AR Smart Glasses and Socially Visible Sensing                                                                                                                                                                                                                                                                                                                     | Master’s thesis, 2026                     |
| – Used a scenario-based experiment with 369 prospective users to compare outward recording, eye tracking, and AI analysis; wearer stigma was the only resistance factor directly associated with lower intention to use.                                                                                                                                          |                                           |
| – Third study in a programme of work on AR smart glasses, alongside the hardware-based proactive-AI and AI-companion experiments below.                                                                                                                                                                                                                           |                                           |
| See, Sense, Spark — Proactive AI for Creative Research Ideation                                                                                                                                                                                                                                                                                                   | SIGGRAPH Asia 2025 Poster                 |
| – Built a proactive AI system on XREAL Air 2 Pro. The poster study (analytic N = 17) found higher interest-driven curiosity with fine-grained profiles (p = .016, d = 1.30). The CHI 2027 follow-up (N = 26; 24 complete logs) found clear differences in AI-generated content, with no detectable differences in participants’ question breadth or self-reports. |                                           |
| No Time to Say Goodbye — Termination and Emotional Loss in AI Companions                                                                                                                                                                                                                                                                                          | CHI EA 2026                               |
| – Built IRO and ran a between-subjects experiment on emotional loss (N = 26; ANCOVA, p = .011). The follow-up, Goodbyes Matter (CHI 2027, under review; N = 56), found that advance notice reduced immediate loss without reducing immersion.                                                                                                                     |                                           |
| ThinkCap — Critical Reading Interface                                                                                                                                                                                                                                                                                                                             | Independent Prototype in Development      |
| – Prototyped a browser extension, native macOS overlay, and BLE firmware for line focus, critical marking, annotation, and behavioural logging; physical integration and evaluation remain in development.                                                                                                                                                        |                                           |
| Phodong: AI-Based Phygital Story Creation for Children                                                                                                                                                                                                                                                                                                            | HCI Korea 2026 Creative Award Grand Prize |
| – Developing a SwiftUI/Firebase system that turns child-discovered objects into personalised stories and traces each object from discovery through story inclusion and child response.                                                                                                                                                                            |                                           |
| Progressive AR Sound Cues                                                                                                                                                                                                                                                                                                                                         | Research System in Development            |
| – Implemented a DHH-informed Unity/XREAL package with separated event sources, cue policies, head-pose logs, and counterbalanced trials; device validation and participant studies remain pending.                                                                                                                                                                |                                           |

SELECTED TECHNICAL DEVELOPMENT

|                                                                                                                                                                                              |                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 3D Gaussian Splatting Reconstruction and Viewer                                                                                                                                              | WebGL / GLSL, 2025            |
| – Built a COLMAP/3D Gaussian Splatting workflow and reimplemented forward rasterisation, alpha blending, depth sorting, and camera-aware optimisation in JavaScript and GLSL.                |                               |
| Neural Models for Visual and Text Classification                                                                                                                                             | PyTorch, 2026                 |
| – Evaluated an MLP for Fashion-MNIST and a padding-aware bidirectional LSTM for IMDb sentiment classification, comparing augmentation, regularisation, pooling, and training configurations. |                               |
| Interactive Graphics and Geometry Processing                                                                                                                                                 | Three.js / WebGL / GLSL, 2025 |
| – Implemented articulated animation and interactive graphics in Three.js, plus half-edge mesh processing, Catmull–Clark subdivision, and derivative-based texture filtering in GLSL.         |                               |

AWARDS, EXHIBITIONS, AND CREATIVE WORK

|                                                                                             |          |
|---------------------------------------------------------------------------------------------|----------|
| HCI Korea 2026 Creative Award · Grand Prize for Phodong                                     | 2026     |
| Selected Team, 2026 Promising Student Startup Team 300+ · Growth Track                      | 2026     |
| GS SHOP Content Leader Scholarship · 13th Cohort                                            | Jan 2025 |
| SKKU Summa Cum Laude · Sungkyunkwan University                                              | Feb 2025 |
| Best Award · 2nd Joint Academic Festival of Convergence Colleges                            | Aug 2024 |
| Best Presentation Award · Cybercommunication Society Spring Conference                      | May 2024 |
| Ars Electronica 2024 · Miracle on the Han River, co-creator and exhibitor, Linz             | 2024     |
| HCI Korea 2024 Creative Award Proceedings · Published contribution                          | 2024     |
| Grand Prize · 2nd Culture and Technology Convergence Major Day Capstone Project Competition | Dec 2023 |

LEADERSHIP AND ACADEMIC SERVICE

|                                                                                                                                                   |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------|
| President, 4th Student Council CNT:role, Culture and Technology Convergence                                                                       | 2023 |
| Led student community programmes at Sungkyunkwan University and helped launch the first joint academic festival across five convergence colleges. |      |

RESEARCH SKILLS

|                      |                                                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming and data | Python; R; Swift; JavaScript; TypeScript; NumPy; pandas                                                                                                                          |
| AI, XR, and graphics | PyTorch; scikit-learn; Unity; Unreal Engine; XREAL; SwiftUI; Firebase; WebGL                                                                                                     |
| Research methods     | Controlled experiments; preregistration; interaction-log analysis; mixed models; permutation/bootstrap inference; SEM; qualitative coding                                        |
| Research areas       | Human–AI interaction; AI contribution and appropriation; proactive AI; agency and metacognition; AR/XR; trace-based evaluation                                                   |
| Languages            | Korean (native); Chinese (proficient — six years of residence in China); English (academic reading, writing, and research communication); Finnish (beginner; currently studying) |

REFERENCES

Prof. Changjun Lee · M.S. advisor · Sungkyunkwan University · changjunlee@skku.edu